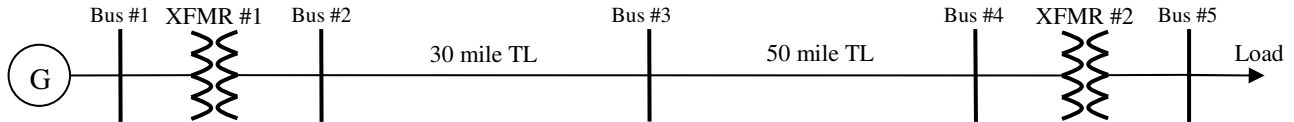


Power system transformers are special interconnect components used to perform voltage level conversion in power systems, while operating at high energy levels and with very high efficiency. Power system transformers that raise the voltage level are called step-up transformers, and power system transformers that lower the voltage level are called step-down transformers. Both step-up and step-down power transformers are used and are integral components in the design and operation of real-world power systems.

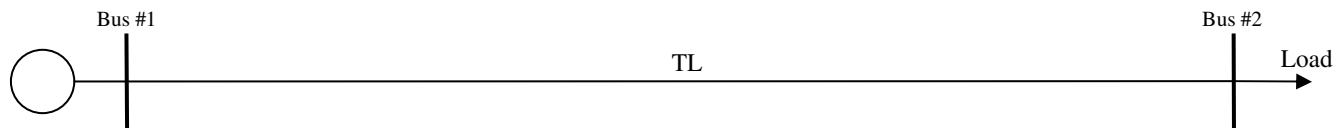
The main reason transformers were invented was to create high voltage levels for power system transmission lines. The example one line diagram below graphically illustrates this application of the transformer. Obviously in this example, XFMR #1 is a step-up transformer, and XFMR #2 is a step-down transformer.



It is desirable to create high voltage levels for power transmission because the same amount of power can be transmitted with a lower current if at a high voltage level compared to a low voltage level. A lower current through the transmission line conductors means that there will be less I^2R real power loss in the conductors, and the transmission line and power system will therefore operate more efficiently. Also lower current through the transmission line conductors means that there will be less voltage drop from one end of a transmission line to the other. Also lower current through transmission line conductors mean that smaller conductors can be used which lowers the material and construction cost of transmission lines.

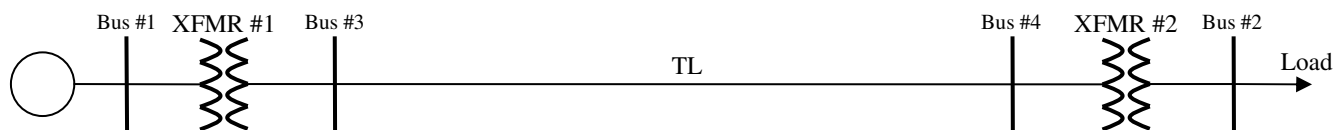
Therefore there are many benefits to operating transmission lines at higher voltage levels in power systems, and these benefits are all made possible by the power transformer. To illustrate this point numerically, two different simple 1 ϕ power systems and some simple calculations are shown below.

System A: Consists of a source, a transmission line with total series impedance of $10+j100\Omega$, and a load which must operate at 25KV and consume 1MW & 1MVAR



$$\begin{aligned}\bar{V}_2 &= 25\angle 0^\circ \text{ KV}, \quad \bar{I}_2 = 56.569\angle -45^\circ \text{ A} \\ \bar{V}_1 &= \bar{V}_2 + \bar{Z}_{TL}\bar{I}_2 = 29.6196\angle 0^\circ \text{ KV}, \quad \bar{I}_1 = \bar{I}_2 = 56.569\angle -45^\circ \text{ A} \\ P_{TL-loss} &= \text{Re}\{\bar{V}_1\bar{I}_1^*\} - \text{Re}\{\bar{V}_2\bar{I}_2^*\} = 32 \text{ KW}\end{aligned}$$

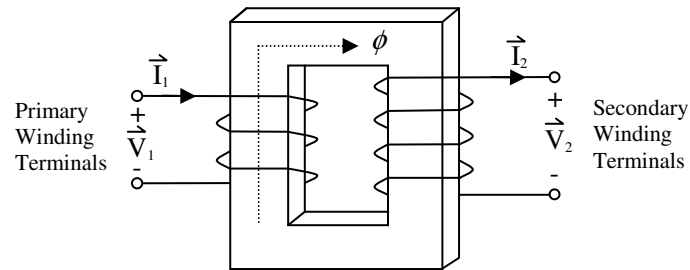
System B: Same as the first system but with transformers on each end of the transmission line to operate the transmission line at a nominal voltage level of 250KV. Assume that these two transformers are both ideal (lossless), having step-up and step-down voltage ratios 25KV/250KV and 250KV/25KV.



$$\begin{aligned}\bar{V}_2 &= 25\angle 0^\circ \text{ KV}, \quad \bar{I}_2 = 56.569\angle -45^\circ \text{ A}, \quad \bar{V}_4 = 250\angle 0^\circ \text{ KV}, \quad \bar{I}_4 = 5.6569\angle -45^\circ \text{ A} \\ \bar{V}_3 &= \bar{Z}_{TL}\bar{I}_4 + \bar{V}_4 = 250.4403\angle 0.082^\circ \text{ KV}, \quad \bar{I}_3 = \bar{I}_4 = 5.6569\angle -45^\circ \text{ A} \\ \bar{V}_1 &= 25.0440\angle 0.082^\circ \text{ KV}, \quad \bar{I}_1 = 56.569\angle -45^\circ \text{ A} \\ P_{TL-loss} &= \text{Re}\{\bar{V}_3\bar{I}_3^*\} - \text{Re}\{\bar{V}_4\bar{I}_4^*\} = 320 \text{ W}\end{aligned}$$

Notice the difference in the calculated transmission line power loss between the two systems. By operating the transmission line at ten times higher voltage, the transmission line power loss is reduced by a factor of 100!

The physical construction of a power transformer is relatively simple, considering the function it performs in power systems. It consists of two coils wrapped around a common magnetic core, as shown in the diagram below of a 1ϕ transformer. The two coils are electrically isolated (not connected) but magnetically coupled.



The two coils are commonly referred to as the **primary winding** and the **secondary winding**. The primary winding is assumed to be the input side of the transformer, and the secondary winding is assumed to be the output side. Notice that the voltage/current sign conventions in the diagram above concur with this assumption.

What makes a transformer step-up or step-down is the relative number of turns on the two windings. An important rule to remember is that **the side of any transformer with more turns on its winding is always the high voltage side**. A special quantity called the TTR (Transformer Turns Ratio) is defined to numerically express the relative number of turns on the two windings. The **TTR is defined as the number of primary winding turns divided by the number of secondary winding turns**. With this definition, a step-up transformer will have a TTR less than 1, and a step-down transformer will have a TTR greater than 1.

The terms **step-up** and **step-down** used to describe power transformers always refers specifically to the primary-secondary voltage behavior of a transformer, and never to the transformer currents. Due to the principle of conservation of energy, the currents in the primary and secondary windings of a transformer will always have the opposite behavior of the voltages. For example in a step-up transformer, the secondary side voltage is larger than the primary side voltage, but the secondary side current is lower than the primary side current.

Mathematical modeling of the detailed physical construction and operation of a transformer is a complicated subject that is best handled in electromagnetics and electric machinery courses. In a power system analysis course, we are more concerned with doing power system modeling and calculations. Therefore the following transformer equivalent circuit models will just be defined for our use and will not be mathematically proven.

Single Phase Power Transformer Equivalent Circuit and Input/Output Equations	
	$\begin{cases} a = \text{TTR} = V_{1\text{rated}} / V_{2\text{rated}} \\ \bar{Z}_x = R_1 + jX_1 \\ \quad = (r_1 + jx_1) + a^2 (r_2 + jx_2) \end{cases} \quad \begin{cases} \bar{V}_2 = \frac{1}{a} (\bar{V}_1 - \bar{Z}_x \bar{I}_1) \\ \bar{I}_2 = a \bar{I}_1 \end{cases}$
Three Phase Power Transformer Per Phase Equivalent Circuit and Input/Output Equations	
	$\begin{cases} a = \text{TTR} = V_{LL1\text{rated}} / V_{LL2\text{rated}} \\ \bar{Z}_x = R_1 + jX_1 \\ \quad = (r_1 + jx_1) + a^2 (r_2 + jx_2) \\ \quad \text{divide } (r + jx) \text{ value by 3} \\ \quad \text{if the side is } \Delta\text{-connected} \end{cases} \quad \begin{cases} \bar{V}_2 = \left(\frac{1}{a} \angle \phi \right) (\bar{V}_1 - \bar{Z}_x \bar{I}_1) \\ \bar{I}_2 = (a \angle \phi) \bar{I}_1 \end{cases}$ $\begin{cases} \phi = 0^\circ \text{ if } Y\text{-}Y \text{ or } \Delta\text{-}\Delta \text{ or } \Delta_{cba}\text{-}\Delta_{cba} \\ \quad = -30^\circ \text{ if } Y\text{-}\Delta \text{ or } \Delta_{cba}\text{-}Y \\ \quad = +30^\circ \text{ if } Y\text{-}\Delta_{cba} \text{ or } \Delta\text{-}Y \end{cases} \quad \begin{cases} \phi = -60^\circ \text{ if } \Delta_{cba}\text{-}\Delta \\ \quad = +60^\circ \text{ if } \Delta\text{-}\Delta_{cba} \end{cases}$

The circuit parameters in the transformer equivalent circuits each model a specific physical characteristic or phenomenon in the transformer. The “ a ” parameter in the transformer model is the TTR of the transformer, and it models the basic step-up or step-down behavior of the transformer. The Z_x parameter models the transformer winding resistances and inductance effects. The ϕ parameter in the 3ϕ transformer equivalent circuit models a special phase shift phenomenon that is unique to 3ϕ transformers and does not exist in 1ϕ transformers.

The circuit parameters in the transformer model are all calculated from the specifications and ratings provided by the transformer manufacturer. Successfully doing this requires an understanding of the transformer specifications and definitions of the rules required to correctly interpret the specifications. The standard minimal information found in a typical transformer specification and how to calculate the transformer model circuit parameters from this information are given below.

Minimal Information Given in Transformer Specifications

1. Rated Maximum Apparent Power Output Capability (S_{rated} or $S_{3\phi\text{rated}}$)
2. Rated Nominal Winding Voltages ($V_{1\text{rated}}/V_{2\text{rated}}$ or $V_{LL1\text{rated}}/V_{LL2\text{rated}}$)
3. Winding Impedances (r_1, x_1, r_2, x_2) or (R_1 and X_1) or (Z_x in % and X/R)
4. Three Phase Connection Scheme (Y or Δ or Δ_{CBA}) of Each Side

The “ a ” parameter or TTR in the transformer model equivalent circuit is calculated using the equations below.

$$\text{for } 1\phi \text{ XFMRs: } a = \text{TTR} = \frac{V_{1\text{rated}}}{V_{2\text{rated}}} \qquad \text{for } 3\phi \text{ XFMRs: } a = \text{TTR} = \frac{V_{LL1\text{rated}}}{V_{LL2\text{rated}}}$$

If the winding impedances are given in (r_1, x_1, r_2, x_2) format with units Ω , then the equivalent circuit impedance “ Z_x ” is calculated using the equations below.

$$\text{for } 1\phi \text{ XFMRs: } \bar{Z}_x = (r_1 + jx_1) + a^2(r_2 + jx_2)$$

$$\text{for } 3\phi \text{ XFMRs: } \bar{Z}_x = (r_1 + jx_1) + a^2(r_2 + jx_2) \quad \text{where } (r_x + jx_x) \text{ is divided by 3 if the side is } \Delta \text{ connected}$$

If the winding impedances are given in (r_1, x_1, r_2, x_2) format with units of % instead of Ω , then the following special conversion is required to obtain the equivalent circuit impedance “ Z_x ” in units of Ω .

$$\text{for } 1\phi \text{ XFMRs: } \bar{Z}_x = \left(\frac{r_1 + jx_1 \%}{100} \right) \left[\frac{(\text{rated primary side KV})^2}{(\text{rated MVA})} \right] + a^2 \left(\frac{r_2 + jx_2 \%}{100} \right) \left[\frac{(\text{rated secondary side KV})^2}{(\text{rated MVA})} \right]$$

$$\text{for } 3\phi \text{ XFMRs: } \bar{Z}_x = \left(\frac{r_1 + jx_1 \%}{100} \right) \left[\frac{(\text{rated primary side KV})^2}{(\text{rated MVA})} \right] + a^2 \left(\frac{r_2 + jx_2 \%}{100} \right) \left[\frac{(\text{rated secondary side KV})^2}{(\text{rated MVA})} \right]$$

again the $(r_x + jx_x)$ terms are divided by 3 if the side is Δ connected

If the winding impedances are given in (R_1, X_1) format with units Ω , then the equivalent circuit impedance “ Z_x ” is calculated using the equation below.

$$\bar{Z}_x = R_1 + jX_1 \quad \text{for both } 1\phi \text{ and } 3\phi \text{ XFMRs}$$

If the winding impedances are given in (R_1, X_1) format with units of % instead of Ω , then the following special per unit conversion is required to obtain the equivalent circuit impedance “ Z_x ” in units of Ω .

$$\bar{Z}_x = \left(\frac{R_1 + jX_1 \text{ in } \%}{100} \right) \left[\frac{(\text{rated primary side KV})^2}{(\text{rated MVA})} \right] \quad \text{for both } 1\phi \text{ and } 3\phi \text{ XFMRs}$$

If the winding impedances are given in (Z_x in % and X/R) format, then the following special per unit conversion is required to obtain the equivalent circuit impedance “ Z_x ” in units of Ω .

$$\bar{Z}_x = \left(\frac{Z_x \text{ in } \%}{100} \right) \left[\frac{(\text{rated primary side KV})^2}{(\text{rated MVA})} \right] \angle \tan^{-1}(X/R) \quad \text{for both } 1\phi \text{ and } 3\phi \text{ XFMRs}$$

Next several examples will be presented that show how to calculate the transformer equivalent circuit model parameters from the transformer specifications and also how to perform basic transformer voltage/current calculations using the transformer model equations.

Example #1:

A 1 ϕ transformer has the specifications: 50KVA, 480V/8KV, $r_1=0.01\Omega$, $x_1=0.11\Omega$, $r_2=0.21\Omega$, $x_2=2.85\Omega$. A 482.375V source is connected to the primary side of this transformer, and a load is connected to the secondary side. The source is observed to supply 39.6285KVA at a 0.925877 lagging power factor. Calculate the voltage and current on the secondary side of the transformer and the equivalent impedance of the load.

$$a = \text{TTR} = \frac{V_{1\text{rated}}}{V_{2\text{rated}}} = \frac{480}{8000} = 0.06$$

$$\bar{Z}_x = (r_1 + jx_1) + a^2(r_2 + jx_2) = (0.01 + j0.11) + a^2(0.21 + j2.85) = 0.12074 \angle 84.889^\circ \Omega$$

$$V_s = 482.375 \text{ V}, \quad S_s = 39.6285 \text{ KVA}, \quad I_s = S_s/V_s = 82.1529 \text{ A}, \quad \theta_s = +\cos^{-1} 0.925877 = 22.199^\circ$$

$$\bar{V}_s = 482.375 \angle 0^\circ \text{ V}, \quad \bar{I}_s = 82.1529 \angle -22.199^\circ \text{ A}$$

$$\bar{V}_2 = \frac{1}{a}(\bar{V}_s - \bar{Z}_x \bar{I}_s) = 7965.0890 \angle -1.057^\circ \text{ V}, \quad \bar{I}_2 = a \bar{I}_s = 4.929 \angle -22.199^\circ \text{ A}$$

$\therefore$ the load voltage and current are 7.9651 KV, 4.929 A

and the equivalent load impedance is $\bar{Z}_L = \bar{V}_2 / \bar{I}_2 = 1615.9077 \angle 21.142^\circ \Omega$

Notice in the previous example that the actual operating voltages of the transformer windings are different from the rated voltages. This is an important thing to remember in transformer modeling and calculations. **One should never assume that the actual operating voltages or output power of a transformer are equal to the rated values in the transformer specifications, unless specifically told to make that assumption.**

Example #2:

A 3 ϕ transformer is connected Y- Δ and has the specifications: 175KVA, 13.8KV/480V, $r_1=0.33\Omega$, $x_1=3.23\Omega$, $r_2=0.03\Omega$, $x_2=0.12\Omega$. A source is connected to the primary side of this transformer, and a load is connected to the secondary side. The load is observed to operate at 475V, 86A, and a 0.901237 lagging power factor. Calculate the source voltage, current, and power factor.

$$a = \text{TTR} = \frac{V_{LL1\text{rated}}}{V_{LL2\text{rated}}} = \frac{13,800}{480} = 28.75, \quad \phi = -30^\circ (\text{Y-}\Delta)$$

$$\bar{Z}_x = (r_1 + jx_1) + a^2 \left(\frac{r_2 + jx_2}{3} \right) = (0.33 + j3.23) + a^2 \left(\frac{0.03 + j0.12}{3} \right) = 37.269519 \angle 76.675^\circ \Omega$$

$$V_{LL2} = 475 \text{ V}, \quad I_{L2} = 86 \text{ A}, \quad pf_2 = 0.901237 \text{ lagging} \rightarrow \theta_2 = +\cos^{-1}(pf_2) = 25.679^\circ$$

$$\bar{V}_2 = (V_{LL2} / \sqrt{3}) \angle 0^\circ = 274.24138 \angle 0^\circ \text{ V}, \quad \bar{I}_2 = I_{L2} \angle (0^\circ - \theta_2) = 86 \angle -25.679^\circ \text{ A}$$

$$\bar{I}_2 = (a \angle \phi) \bar{I}_s \rightarrow \bar{I}_s = \frac{\bar{I}_2}{(a \angle \phi)} = 2.9913 \angle 4.321^\circ \text{ A}$$

$$\bar{V}_2 = (a^{-1} \angle \phi)(\bar{V}_s - \bar{Z}_x \bar{I}_s) \rightarrow \bar{V}_s = \frac{\bar{V}_2}{(a^{-1} \angle \phi)} + \bar{Z}_x \bar{I}_s = 7955.1275 \angle 30.624^\circ \text{ V}$$

$$\theta_s = \alpha_s - \beta_s = 30.624^\circ - 4.321^\circ = 26.303^\circ$$

∴ the source voltage, current, and power factor are:

$$V_{LLS} = \sqrt{3} \cdot |\bar{V}_S| = 13.7787 \text{ KV}, \quad I_{LS} = |\bar{I}_S| = 2.991 \text{ A}, \quad pf_s = \cos \theta_s = 0.896461 \text{ lagging}$$

Example #3:

A 3 ϕ transformer is connected Δ -Y and has the specifications: 100MVA, 161KV/13.2KV, 10% Z, 42 X/R. A source is connected to the primary side of this transformer, and a load is connected to the secondary side. The source is observed to operate at 160.375KV, 387A, and a 0.89534 leading power factor. Calculate the voltage, current, and power factor on the secondary side of the transformer.

$$a = \text{TTR} = \frac{V_{LL1\text{rated}}}{V_{LL2\text{rated}}} = \frac{161}{13.2} = 12.19697, \quad \phi = +30^\circ (\Delta\text{-Y})$$

$$\bar{Z}_x = \left(\frac{Z_x \text{ in } \%}{100} \right) \left[\frac{(\text{rated primary side KV})^2}{(\text{rated MVA})} \right] \angle \tan^{-1}(X/R) = \left(\frac{10}{100} \right) \left(\frac{161^2}{100} \right) \angle \tan^{-1}(42) = 25.921 \angle 88.636^\circ \Omega$$

$$V_{LL1} = 160.375 \text{ KV}, \quad I_{L1} = 387 \text{ A}, \quad pf_1 = 0.89534 \text{ leading} \rightarrow \theta_1 = -\cos^{-1}(pf_1) = -26.448^\circ$$

$$\bar{V}_1 = (V_{LL1} / \sqrt{3}) \angle 0^\circ = 92,592.549 \angle 0^\circ \text{ V}, \quad \bar{I}_1 = I_{L1} \angle (0^\circ - \theta_1) = 387 \angle 26.448^\circ \text{ A}$$

$$\bar{V}_2 = (a^{-1} \angle \phi)(\bar{V}_S - \bar{Z}_x \bar{I}_S) = 7974.977 \angle 24.641^\circ \text{ V}$$

$$\bar{I}_2 = (a \angle \phi) \bar{I}_S = 4720.227 \angle 56.448^\circ \text{ A}$$

$$\theta_2 = \alpha_2 - \beta_2 = 24.641^\circ - 56.448^\circ = -31.807^\circ$$

∴ the voltage, current, and power factor on the transformer secondary side are:

$$V_{LL2} = \sqrt{3} \cdot |\bar{V}_2| = 13.8131 \text{ KV}, \quad I_{L2} = |\bar{I}_2| = 4720.23 \text{ A}, \quad pf_2 = \cos \theta_2 = 0.849826 \text{ leading}$$

Percent loading is very important and useful power transformer calculation. It is defined as the ratio in percent of the actual power output of a transformer to the rated maximum power output capability of the transformer. It is a measure of what percentage of a transformer's maximum power output capability that the transformer is actually operating at. Apparent power must be used in transformer percent loading calculations since the transformer maximum power rating is always given using apparent power. The percent loading of a transformer is mathematically defined below.

$$\%L = \left(\frac{\text{actual apparent power output}}{\text{rated apparent power output}} \right) \times 100 = \left(\frac{S_2}{S_{\text{rated}}} \right) \times 100 \quad \text{or} \quad \%L = \left(\frac{S_{3\phi 2}}{S_{3\phi \text{rated}}} \right) \times 100$$

For the previous Example #1, the transformer percent loading is calculated as shown below.

$$S_2 = V_2 I_2 = (7965.089)(4.929) = 39,261.304 \text{ VA} = 39.2613 \text{ KVA}$$

$$\%L = \left(\frac{S_2}{S_{\text{rated}}} \right) \times 100 = \left(\frac{39.2613 \text{ KVA}}{50 \text{ KVA}} \right) \times 100 = 78.523\%$$

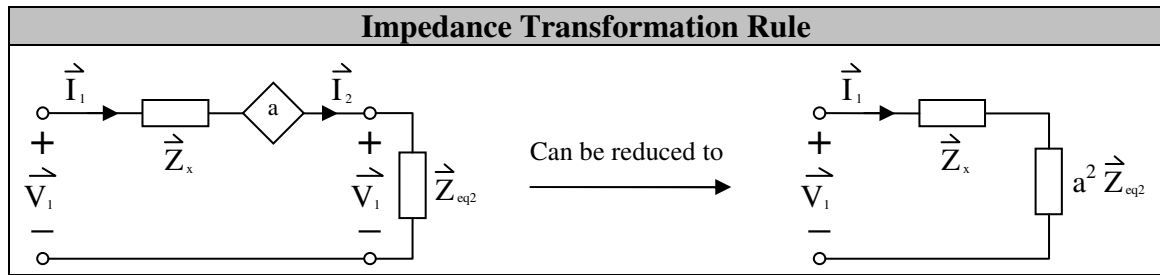
For the previous Example #3, the transformer percent loading is calculated as shown below.

$$S_{3\phi 2} = 3 \cdot S_2 = 3(V_2 I_2) = 3(7974.977)(4720.227) = 112,931,114.97 \text{ VA} = 112.931115 \text{ MVA}$$

$$\%L = \left(\frac{S_{3\phi 2}}{S_{3\phi \text{rated}}} \right) \times 100 = \left(\frac{112.931115 \text{ MVA}}{100 \text{ MVA}} \right) \times 100 = 112.931\% \text{ (overloaded)}$$

When analyzing power systems containing transformers using 5-step PSA, a circuit analysis of the power system equivalent circuit must be done. The special diamond symbol in the transformer model must be handled in a special way when doing circuit analysis since it is not a standard type of circuit element like a source or

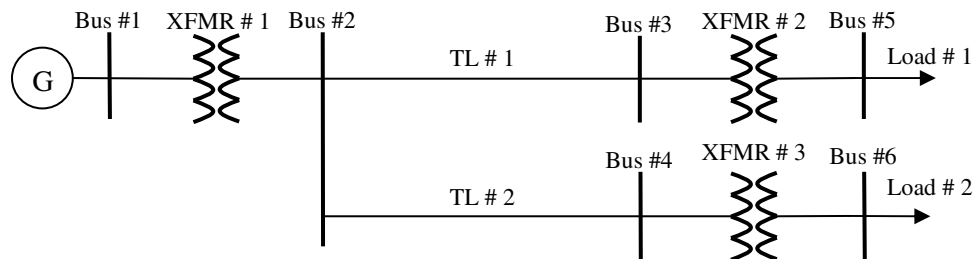
impedance. An important principle called the **impedance transformation rule** is defined and used to simplify the analysis of equivalent circuits containing the transformer model. This special rule is defined below.



The impedance transformation rule should be thought of as a circuit analysis tool to reduce power system equivalent circuits containing the transformer model. The example problem below illustrates this point.

Example #4:

The one line diagram and component specifications for a 3 ϕ power system are given below. Calculate the load voltages.

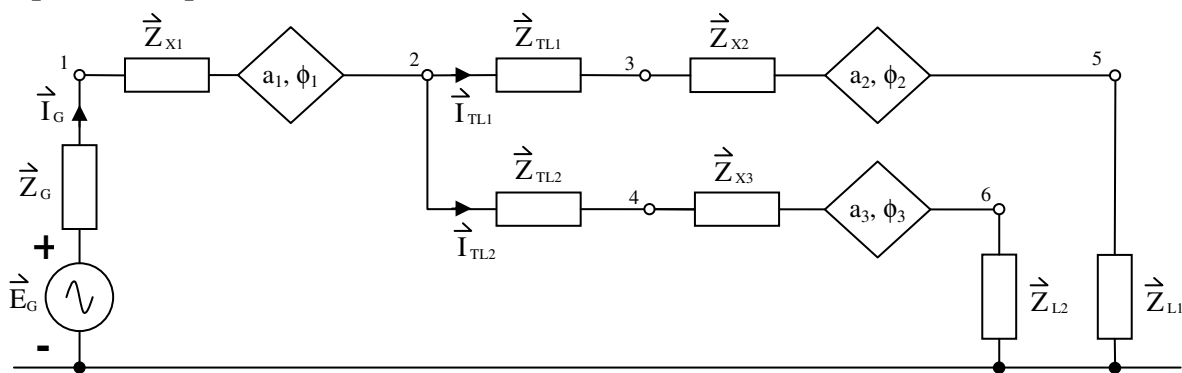


- Generator:** 500KVA, 4160V, Y, 21% X, 162 X/R, no-load voltage 4325V
XFMR #1: 500KVA, 4160V/23KV, Y- Δ , 3.1% Z, 4.85 X/R
XFMR #2: 100KVA, 23KV/480V, Δ -Y, 2.1% Z, 3.25 X/R
XFMR #3: 200KVA, 23KV/208V, Δ -Y, 3.2% Z, 3.69 X/R
TL #1: 3-conductor power cable with total per conductor impedance $2.82\angle 30^\circ \Omega$
TL #2: 3-conductor power cable with total per conductor impedance $5.49\angle 30^\circ \Omega$
Load #1: Δ -connected with per phase impedance $12\angle 65^\circ \Omega$
Load #2: Y-connected with per phase impedance $1\angle 35^\circ \Omega$

Step 1: Visualize

A one line diagram for the system was given.

Step 2: L-N Equivalent Circuit



Step 3: Circuit Parameters

$$X_G = \left(\frac{21\%}{100} \right) \left[\frac{(4.160)^2}{(0.500)} \right] = 7.268352 \, \Omega, \quad R_G = \frac{X_G}{162} = 0.044866 \, \Omega$$

$$\bar{Z}_G = R_G + jX_G = 7.26849 \angle 89.646^\circ \Omega, \quad \bar{E}_G = \left(\frac{4325}{\sqrt{3}} \right) \angle 0^\circ = 2497.0399 \angle 0^\circ V$$

$$\bar{Z}_{L1} = \frac{1}{3} \bar{Z}_{\Delta L1} = 4 \angle 65^\circ \Omega, \quad \bar{Z}_{L2} = \bar{Z}_{YL2} = 1 \angle 35^\circ \Omega$$

$$\bar{Z}_{TL1} = 2.82 \angle 30^\circ \Omega, \quad \bar{Z}_{TL2} = 5.49 \angle 30^\circ \Omega$$

$$a_1 = \frac{4.160}{23}, \quad \phi_1 = -30^\circ, \quad \bar{Z}_{x1} = \left(\frac{3.1\%}{100} \right) \left[\frac{(4.160)^2}{(0.500)} \right] \angle \tan^{-1}(4.85) = 1.072947 \angle 78.350^\circ \Omega$$

$$a_2 = \frac{23}{0.480}, \quad \phi_2 = 30^\circ, \quad \bar{Z}_{x2} = \left(\frac{2.1\%}{100} \right) \left[\frac{(23)^2}{(0.100)} \right] \angle \tan^{-1}(3.25) = 111.09 \angle 72.897^\circ \Omega$$

$$a_3 = \frac{23}{0.208}, \quad \phi_3 = 30^\circ, \quad \bar{Z}_{x3} = \left(\frac{3.2\%}{100} \right) \left[\frac{(23)^2}{(0.200)} \right] \angle \tan^{-1}(3.69) = 84.64 \angle 74.837^\circ \Omega$$

Step 4: Circuit Analysis

$$\bar{Z}_{eq25} = \bar{Z}_{TL1} + \bar{Z}_{x2} + a_2^2 \bar{Z}_{L1} = 9296.384 \angle 65.084^\circ \Omega$$

$$\bar{Z}_{eq26} = \bar{Z}_{TL2} + \bar{Z}_{x3} + a_3^2 \bar{Z}_{L2} = 12,297.835 \angle 35.250^\circ \Omega$$

$$\bar{Z}_{eq2} = \left(\bar{Z}_{eq25}^{-1} + \bar{Z}_{eq26}^{-1} \right)^{-1} = 5475.144 \angle 52.288^\circ \Omega$$

$$\bar{I}_G = \bar{E}_G \left(\bar{Z}_G + \bar{Z}_{x1} + a_1^2 \bar{Z}_{eq2} \right)^{-1} = 13.431 \angle -53.792^\circ A$$

$$\bar{V}_1 = \bar{E}_G - \bar{Z}_G \bar{I}_G = 2418.592 \angle -1.355^\circ V$$

$$\bar{V}_2 = \left(a_1^{-1} \angle \phi_1 \right) \left(\bar{V}_1 - \bar{Z}_{x1} \bar{I}_G \right) = 13,300.404 \angle -31.505^\circ V$$

$$\bar{I}_{TL1} = \bar{V}_2 \left(\bar{Z}_{TL1} + \bar{Z}_{eq3} \right)^{-1} = 1.431 \angle -96.589^\circ A$$

$$\bar{V}_3 = \bar{V}_2 - \bar{Z}_{TL1} \bar{I}_{TL1} = 13,297.103 \angle -31.495^\circ V$$

$$\bar{V}_5 = \left(a_2^{-1} \angle \phi_2 \right) \left(\bar{V}_3 - \bar{Z}_{x2} \bar{I}_{TL1} \right) = 274.219 \angle -1.589^\circ V$$

$$\bar{I}_{TL2} = \bar{V}_2 \left(\bar{Z}_{TL2} + \bar{Z}_{eq4} \right)^{-1} = 1.082 \angle -66.755^\circ A$$

$$\bar{V}_4 = \bar{V}_2 - \bar{Z}_{TL2} \bar{I}_{TL2} = 13294.491 \angle -31.502^\circ V$$

$$\bar{V}_6 = \left(a_3^{-1} \angle \phi_3 \right) \left(\bar{V}_4 - \bar{Z}_{x3} \bar{I}_{TL2} \right) = 119.592 \angle -1.755^\circ V$$

Step 5: Final Results

$$\text{Voltage at Load \#1} = \sqrt{3} \cdot |\bar{V}_5| = 474.961 V$$

$$\text{Voltage at Load \#2} = \sqrt{3} \cdot |\bar{V}_6| = 207.139 V$$

The previous example was for a 3 ϕ power system. The impedance transformation rule works exactly the same way in L-N equivalent circuits for 3 ϕ power systems as it does in equivalent circuits for 1 ϕ power systems. The impedance transformation rule is not affected by phase shift parameter ϕ in the L-N equivalent circuit model for a 3 ϕ transformer. The phase shift parameter ϕ only affects voltage and current calculations done with the transformer model.